

Yoga Pose Corrector

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Problem

Too many injuries in workouts

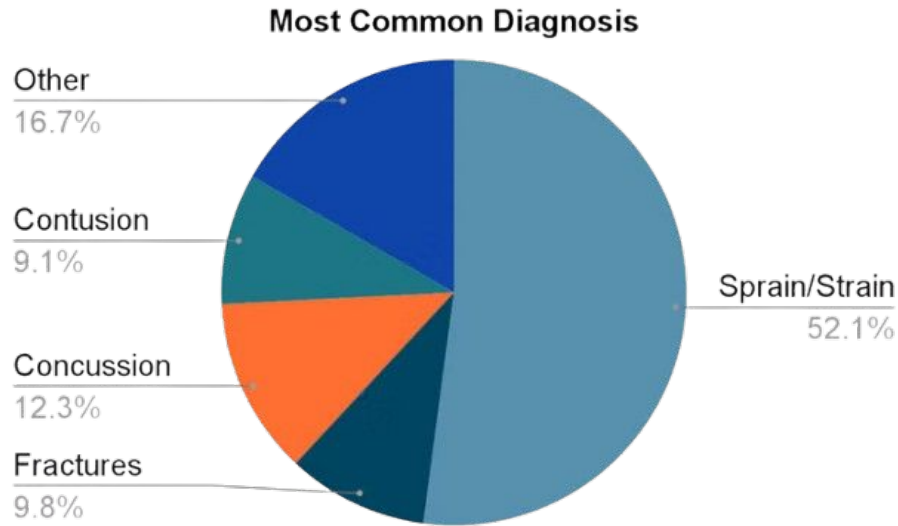
No warmup → Hurt



Solution

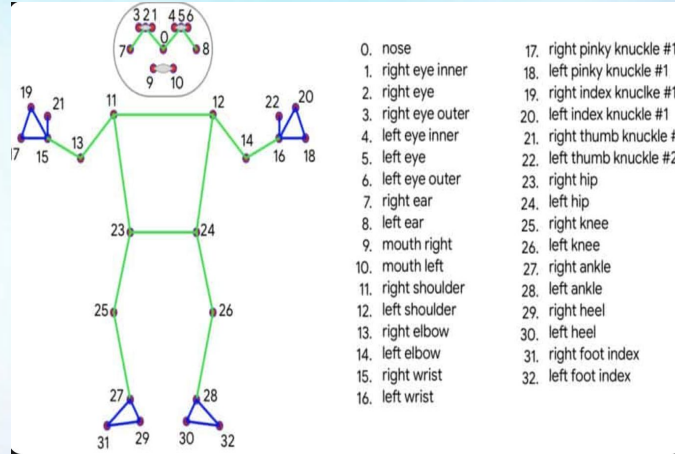
Stretching → Yoga → Physical Activity

Building a real-time yoga pose correcting feedback system based on user video



System Pipeline

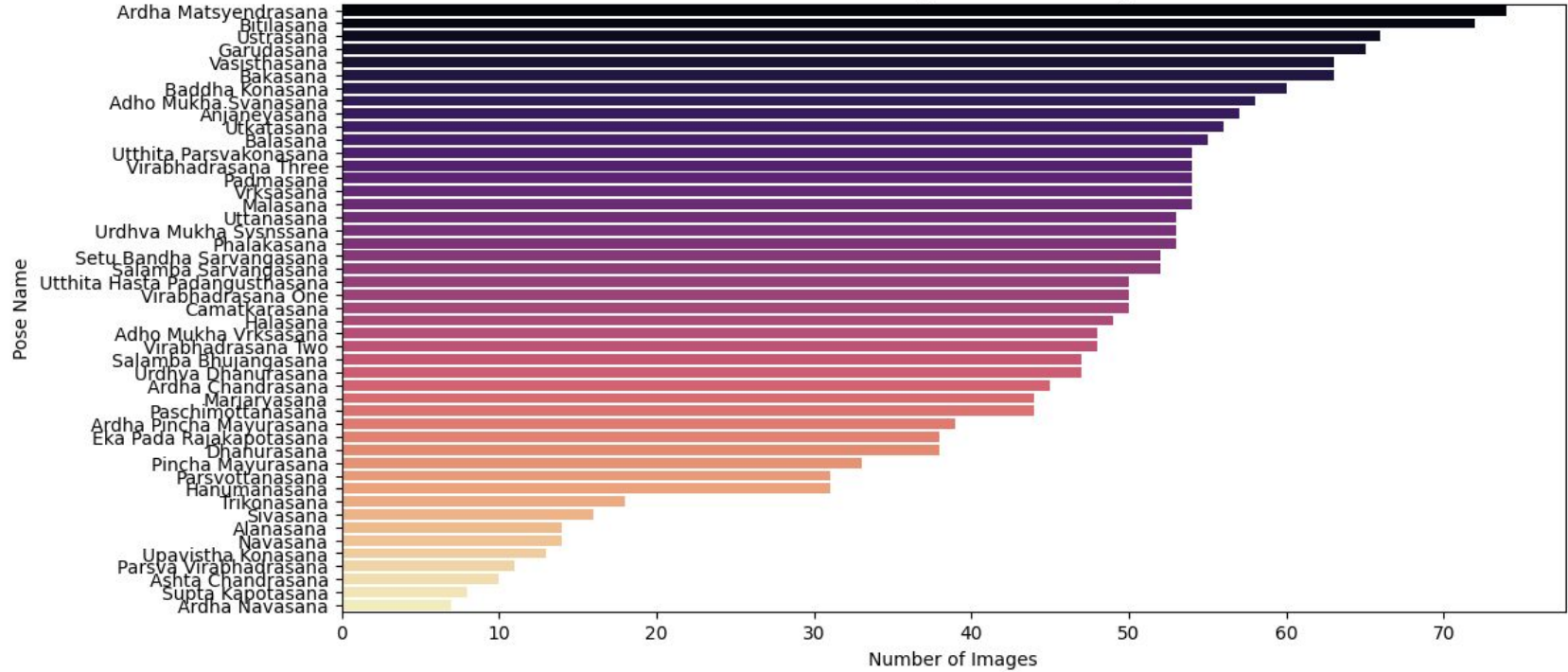
- **Step 1**
 - Use a pretrained model such as **MediaPipe BlazePose** to extract different joint points in the pose.
 - Each of the points will be the features (left elbow, right knee, etc.)
 - Store these in a CSV file and perform **EDA**.
 - Then create a new CSV file with cleaned data.
- **Step 2**
 - Train a neural network on the cleaned data to classify each pose
 - We did train regression, yet decided that neural network was better approach
 - Tune hyperparameters, change dataset cleaning to improve accuracy
- **Step 3:**
 - Manually set angle thresholds between different joints to determine what poses are incorrect
 - Final prediction should
 - Classify the pose
 - Check the angles to detect whether they are correct



Dataset

- ❖ **Source:** We used the Kaggle Yoga Posture Dataset, which contains thousands of images categorized into 40 different yoga poses.
<https://www.kaggle.com/datasets/tr1gg3rtrash/yoga-posture-dataset>
- ❖ **Turning Pixels into Math:** Instead of feeding raw images into the model, we used MediaPipe to extract 33 skeletal landmarks. This turned every image into a list of 132 numerical coordinates (x, y, z, and visibility).
- ❖ **Fixing the Noise:** We filtered out bad data where the camera couldn't see key joints (like hips or knees) to make sure the model only learned from clear, high quality poses.
- ❖ **Leveling the Playing Field:** We normalized the coordinates to a 0.0–1.0 range. This makes it so that the model recognizes a Warrior II whether the person is standing 3 feet or 10 feet away from the camera.
- ❖ **The Split:** We divided the data into training and testing sets to ensure we could prove the model works on photos it has never seen before.

Distribution of Yoga Poses in Training Set



Best Model Developed: Deep Neural Network

- ❖ We built a Deep Dense Neural Network. While CNNs are common for images, a Dense model is actually much faster and more accurate for processing these skeletal landmark coordinates.
- ❖ **Breaking the Bias:** Our first attempt suffered from model collapse, where the brain just guessed the same pose for everything. We fixed this by adding batch normalization and Dropout layers to keep the model balanced.
- ❖ **Smarter Training:** We used a learning rate scheduler. Think of this like fine-tuning a radio, as the model got closer to the right answer, it slowed down its adjustments to find the absolute best accuracy.
- ❖ **Safety First:** We used early stopping, which automatically stops the training the second the model starts overfitting or memorizing the photos instead of learning the poses.
- ❖ **The Final Score:** We hit a **test accuracy of 82.37%**. Since there are 40 different poses, a random guess would only be 2.5%, so our model is performing exceptionally well.

Pose Correction

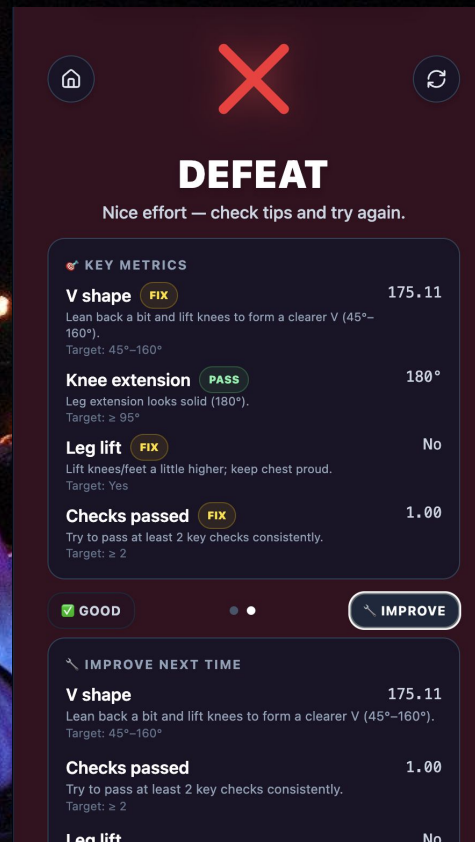
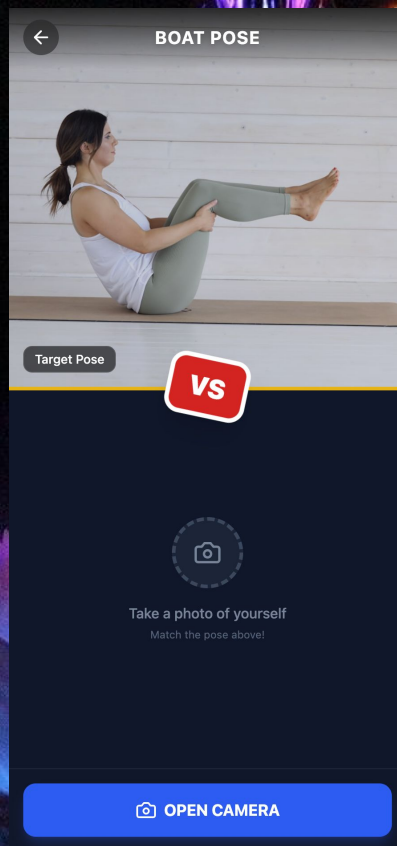
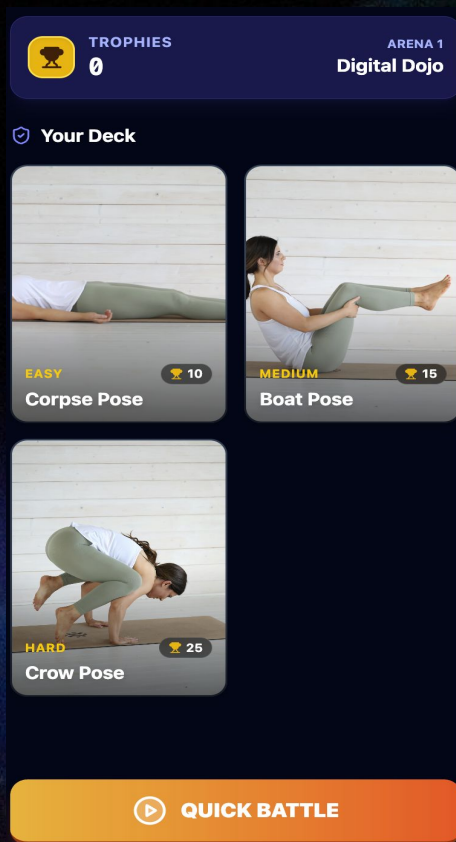
We picked 3 poses for correction in the current version

- ❖ Corpse Pose
- ❖ Boat Pose
- ❖ Crow Pose

Calculated the angle and relative position for each of the joints in the poses and compared with a manually set threshold

Example: Shoulder and knee should be higher than the hips for the boat pose

Web Demo



Limitations and Improvements

- ❖ **Data Augmentation:** To reach 90%+ accuracy, we can programmatically rotate our current landmark data to simulate the user standing at different angles (45° , 90° , etc.), which could further increase the accuracy of our model.
- ❖ **Time Series Analysis (LSTM):** Instead of looking at a single frame, we could look at a three second time frame of movement. This would help the model understand the transition into a pose, not just the final hold that it see and compares the perceived image to.
- ❖ **Correction Threshold Setting:** Add more poses to be corrected and accommodate variations of the poses.
- ❖ **Further Data cleaning:** One method we can use for further feature extraction is removing poses that aren't fully in the frame of the image